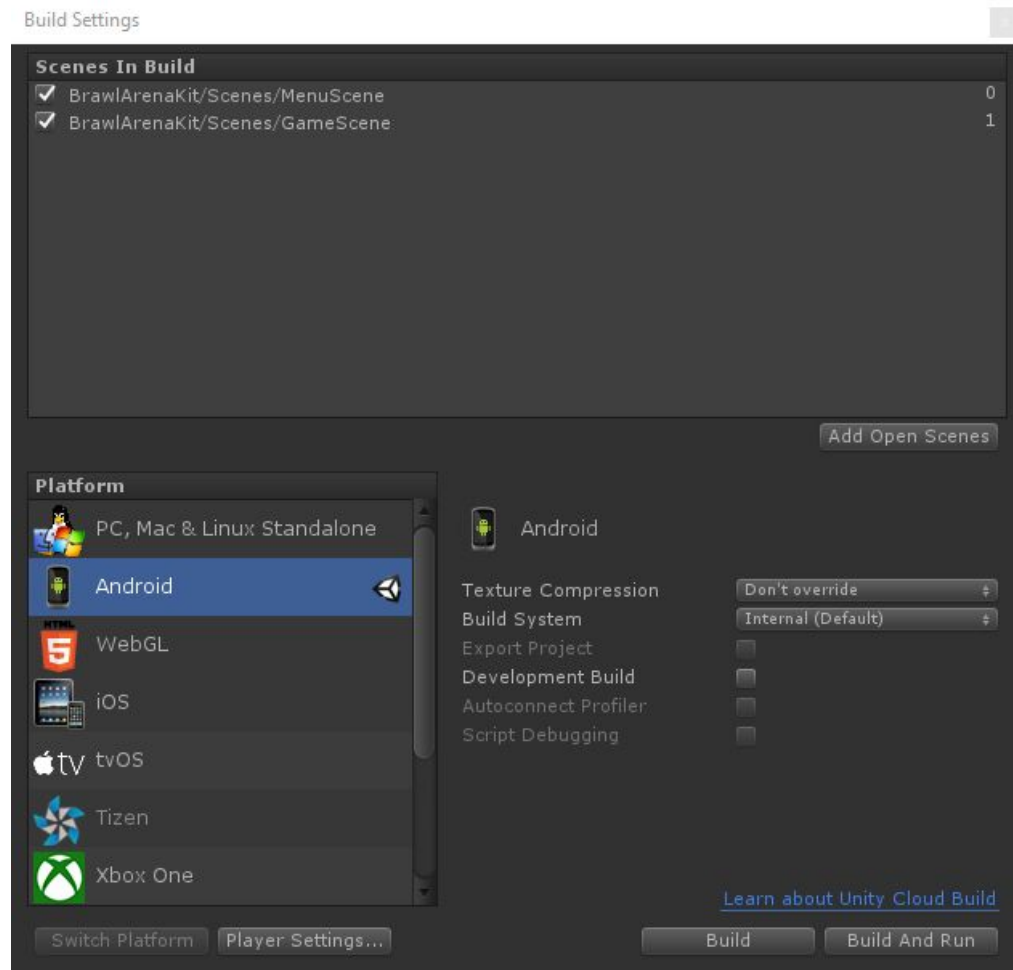


Brawl Arena Base Kit

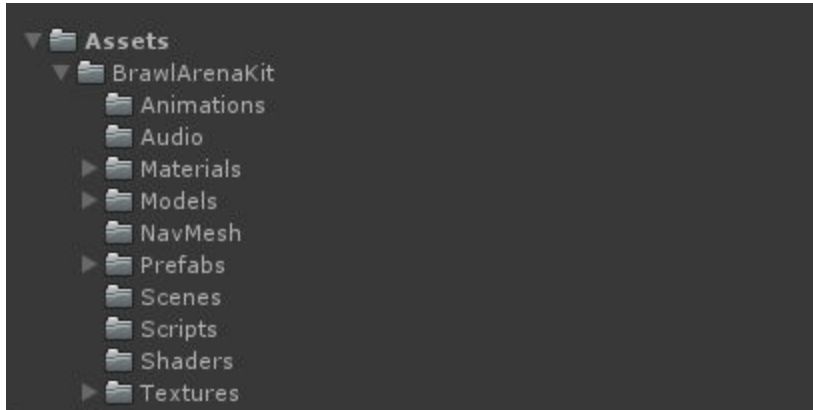
1. Introduction

- Required Unity Version : 5.5.2f1 or Higher
- To setup the project:
 - Import package into Unity
 - Make sure that the following scenes are added in build settings



- [illegible]

2. Contents:



- Animations: contains all animations controllers and animation masks
- Audio: contains audios for bullet fire and impacts
- Materials: contains materials for all characters, environment models & particle materials
- Model: contains all base meshes
- Prefabs: contains prefabs of all environment assets, player characters, AI characters & bullets
- Scenes: contains the menu and game scene
- Scripts: contains all scripts used in projects
- Shaders: contains extra shaders required
- Textures: contains all UI textures, Particles textures, Environment textures & Character textures.

3. Input

Input is provided using two joysticks and one button:

One joystick for moving

One joystick for attacking/aiming

Button for using Ultimate

4. Scripts

- **AIController (virtual parent class)**

Parent script for all AI scripts : all AI scripts are derived from AIController

Start() initializes all values (speed, ammo, layers based on team)

OnTriggerEnter(), OnTriggerExit() keeps track of collisions with grass to keep character invisible

Update() virtual function for parent class: Update function of child class is called based on AI behaviour

DeathReset() resets values on death

AddUltimate() adds ultimate charge when attacking enemy

Children Classes: NinjaAI, WizAI, DjangoAI

- **BulletScript**

Handles all collisions for bullet

OnTriggerEnter(), OnTriggerExit() used by grenade type bullets which deal damage in an area

OnCollisionEnter(), OnCollisionExit() checks collisions: upon collision with walls destroys bullets and instantiate hit particles, upon collision with enemy destroys bullets adds ultimate charge and instantiate hit particles

- **FireMode**

Gun script: decides which type of gun is used : single fire, semi auto, shotgun or grenade type guns

Contains values for bullets which will be instantiated: life, velocity, bullet count, fire rate

Fire() fires bullets based on current gun type: grenades are thrown using physics, shotgun bullets are spread across an angle, semi auto are fired after intervals

- **GameManager**

Handles instantiation and initialization of player character, assigning button and joystick handlers, keeping track of (respawning and killing) all characters in game

Awake() instantiates player and assigns button/joystick handlers to player

Start() populates list of characters in game

KillPlayer() kills a character and starts respawn timer

Update() used to display a timer on screen when main player is dead

Respawn() coroutine to respawn a character

- **HealthController**

manages health of characters/player and HealthBar above character

Start() initializes hp values for given character

OnEnable() resets hp values on respawn

addHealth() heals character

DeductHealth() deducts health and manages character death scenario

Update() used to check if character has not taken damage over time and can self heal

- **PlayerController**

Handles player controls and interactions with environment (does not manage player health : see HealthController)

Start() initializes all values (speed, ammo,), assigns handlers for joysticks

OnTriggerEnter(),OnTriggerExit() keeps track of collisions with grass to keep character invisible

Update() virtual function for parent class: Update function of child class is called based on AI behaviour

DeathReset() resets values on death

AddUltimate() adds ultimate charge when attacking enemy

MarkIsNotAiming(),MarkIsAiming() used for showing/hiding aim UI when fire joystick is held down/dragged

Update() used for moving player, managing aim, refilling ammo

- **UltimateBase(virtual parent class)**

Parent class for ultimate abilities : all ultimate launchers are derive from UltimateBase

Fire() virtual function used by child class to launch current ultimate

Children Classes: PoofUltimate, MachineUltimate, HealUltimate

5. Creating New Weapons/Characters

- **Gun**

Create an empty GamObject and assign the FireMode script to it

Select a GunType from the inspector, to create a new gun type add a new value to the “GunType” enum in the Fire() function add a new case to the switch statement for the new GunType containing the behaviour for the new gun type

From the inspector fill in the values for ultimate charge per hit and other bullet properties

- **Ultimate**

Create a new script derived from UltimateBase:

```
public class NewUltimate : UltimateBase{.....}
```

Implement the virtual Fire() function for when the ultimate is launched add new values for ultimate as required

- **Player Character**

- Create a new character gameobject prefab on the prefab apply the PlayerController Script and HealthController Script
- Add Rigidbody with mass 50 and drag 2
- Add a capsule collider
- Add the animator from animations folder

- Add the StatsCanvas prefab inside the Player prefab
- Add the BulletPivot/ProjectilePivot prefab inside the Player prefab based on gun type
- Add an empty gameobject inside the Player prefab and apply the Gun script on the gameobject (fill in the required values for script)
- Add an empty gameobject inside the Player prefab and apply the required Ultimate Child class script on the gameobject (fill in the required values for script)
- Select the new Player prefab and in the inspector and drop the ForeGround of Health in StatsCanvas on the HealthBar in HealthController Script
- Select the new Player prefab and in the inspector and fill the values of PlayerController Script:
 - Base Speed
 - Base Health
 - Fire Rate
 - Ammo Fill Time
 - Is Throwable : check if using grenade type weapons
 - Gun : Assign child gameobject with GunType script
 - Ultimate : Assign child gameobject with BaseUltimate child script
 - Aim : assign the AimPivot gameobject
 - Assign the required materials for invisible grass normal grass normal player and semi invisible player
 - Player Model : Assign Mesh Renderer of the model
 - Ammo Amount Bar : Assign ForeGround of Ammo in StatsCanvas

- **AI Character**

- Create a new character gameobject prefab on the prefab apply the Required AIController child class and HealthController Script
- Add the StatsCanvas prefab inside the Character prefab

- Add an empty gameobject inside the Character prefab and apply the Gun script on the gameobject (fill in the required values for script)
- Add Rigidbody with iskinematic checked
- Add a capsule collider
- Add the animator from animations folder
- Add a NavMeshAgent with speed 3.5 and Angular Speed 360
- Add an empty gameobject inside the Character prefab and apply the required Ultimate Child class script on the gameobject (fill in the required values for script)
- Select the new Character prefab and in the inspector and drop the ForeGround of Health in StatsCanvas on the HealthBar in HealthController Script
- Select the new Character prefab and in the inspector and fill the values of PlayerController Script:
 - Base Speed
 - Base Health
 - Fire Rate
 - Ammo Fill Time
 - Is Throwable : check if using grenade type weapons
 - Gun : Assign child gameobject with GunType script
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